ZAKKYBOY

How to put games on your ZakkyBoy (ZB-1)

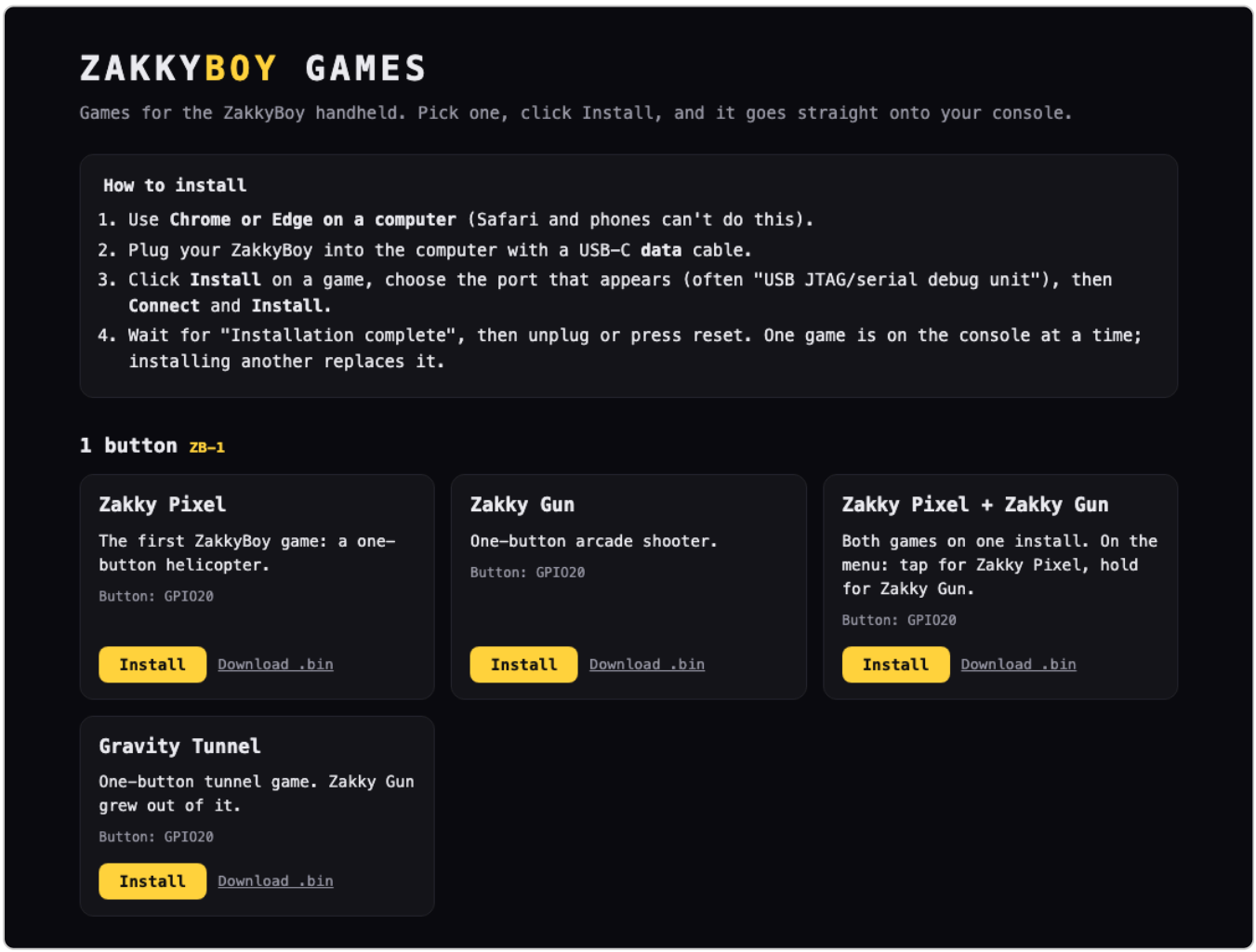
What you need

- Your ZakkyBoy
- A **USB-C data cable**. Some cables only charge; if nothing shows up later, try a different one.
- A **Windows PC or Mac** with **Google Chrome** or **Microsoft Edge**. Safari, phones and iPads can't do this.

You don't need to install any software or make an account.

Installing a game

1. **Open the ZakkyBoy Games page in Chrome or Edge:**
   `zakkyboy313.github.io/zakkyboy-games`
2. **Plug your ZakkyBoy into the computer.**
3. **Find a game and click the yellow Install button.** Games are grouped by how many buttons they need. Your ZB-1 has one button, so pick from the **1 button** section at the top.



4. **Choose your ZakkyBoy from the list that pops up**, then click **Connect**. It's usually called *USB JTAG/serial debug unit*, followed by something like `COM3` on Windows or `cu.usbmodem…` on a Mac.

5. **Click Install.** It asks whether to erase the device first. You can leave this unticked. Ticking it also wipes saved high scores, but it's worth doing if a game ever acts strangely.

6. **Wait for "Installation complete"** (about 20–40 seconds), then unplug the ZakkyBoy or press its reset button. The game starts by itself.

One game at a time. Installing a new game replaces the one already on it. You can switch back whenever you like. If you want two games on one install, use **Zakky Pixel + Zakky Gun**: on its menu, tap the button for Zakky Pixel or hold it for Zakky Gun.

The games

Your ZB-1 plays the 1-button games. The others are there in case you build a ZakkyBoy with more buttons; the pins they need are shown on the games page.

Game	What it is
1 button: ZB-1 (button on GPIO20)	
Zakky Pixel	The first ZakkyBoy game: a one-button helicopter.
Zakky Gun	One-button arcade shooter.
Zakky Pixel + Zakky Gun	Both on one install. Tap for Zakky Pixel, hold for Zakky Gun.
Gravity Tunnel	One-button tunnel game. Zakky Gun grew out of it.
2 buttons: ZB-2 (GPIO20 and GPIO3)	
Zakky Dungeon	Card dungeon: fight monsters, drink potions and pick up weapons from a deck.
Zakky Drift	Drift racer. One button steers left, the other right.
6 buttons	
Blocks	Falling blocks, played upright. Needs an SH1106 screen.
7 buttons	
Jake's Quest	Roguelike: move, shoot and bomb your way through.

New games will appear on the same page as they're finished, so check back now and then.

If something goes wrong

Problem	Try this
There's no Install button, or it says the browser isn't supported	Open the page in Chrome or Edge on a computer.
The pop-up list is empty	Try a different USB-C cable (some only charge) or another USB port. Still nothing? Hold the **BOOT** button on the little board, plug the cable in while holding it, let go, then click Install again.
It stops partway with an error	Unplug, plug back in and try again. If it keeps failing, do the BOOT trick above and tick "erase" this time.
It installed but the screen stays blank	Press reset or unplug and replug. If it's still blank, tell Zak.
The game runs but the button does nothing	Your button is wired to a different pin from the one these games expect (GPIO20). Tell Zak which pin it's on and he'll make a version for it.

Doing it by hand (if the page ever doesn't work)

1. Go to **`github.com/zakkyboy313/zakkyboy-games`**, open the `firmware` folder and click the game's `.bin` file, e.g. `zakky-pixel.bin`. Click the download button (↓) at the top right of the file.
2. In Chrome or Edge, go to **`espressif.github.io/esptool-js`**.
3. Click **Connect** and choose your ZakkyBoy.
4. Set **Flash Address** to `0x0`, choose the `.bin` file you downloaded and click **Program**.
5. When it finishes, unplug the ZakkyBoy or press reset.

Why .bin and not .ino? An `.ino` file is the game's source code, the version people read and edit. Before it can run on the ZakkyBoy it has to be turned into a `.bin` file, which is what you install. That step has already been done for every game on the page, so you never need the `.ino` files.